

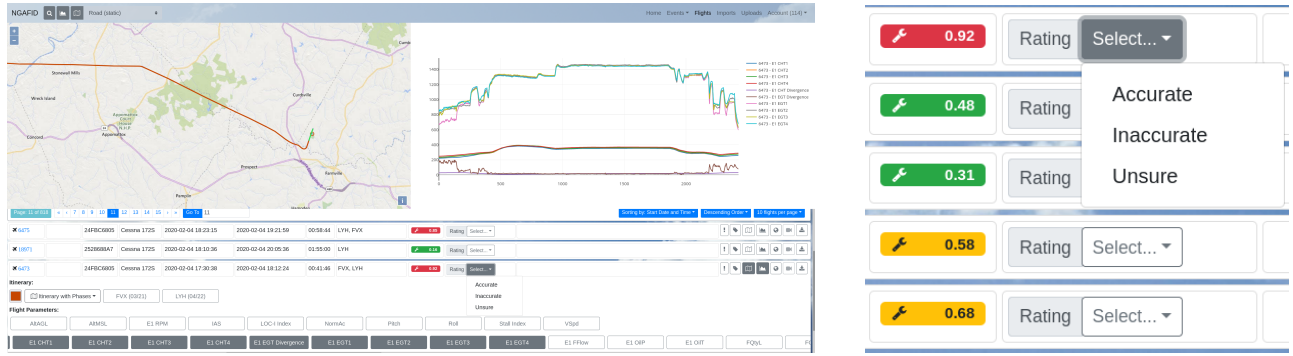


Figure 5: Screenshots showing the integration of the maintenance prediction models into the NGAFID user interface.

were not included, these will be included in the future as they are annotated by domain experts.

NGAFID Deployment and Integration

The NGAFID provides a set of utilities for Flight Data Monitoring (FDM), which allow users to access the per-second time series data and perform various analytics. We added additional functionality to calculate and display the probability that a flight may require maintenance for the Cessna 172S aircraft type for flights exceeding 30 minutes (see Figure 5). This includes a feedback system was created to give users the ability to rate the accuracy of $P(Y_i|X_i)$ using a three-point scale (accurate, inaccurate or unsure), based on their knowledge of aviation and aircraft maintenance. This allows users to provide valuable feedback and labeled data for refining and improving future models.

However, there are infrastructure challenges that need to be addressed before NGAFID can provide real time predictive maintenance alerts to improve safety and reduce costs. The main obstacle is the lack of wireless flight data transmission (WFDT), which is more common in commercial aviation settings. The current data import process for the NGAFID occurs weekly and requires ground crews to manually extract and upload the data. NGAFID partner fleets are in the process of deploying WFDT systems that will allow the NGAFID to perform real time predictive maintenance, as the WFDT systems can upload data immediately after an aircraft lands and returns to the hangar.

Conclusion

We demonstrate the challenging nature of the NGAFID-MC dataset and its value for assessing various MTS approaches. While some datasets exceed NGAFID-MC in terms of datapoints or sequence length, the authors are not aware of any dataset that has both greater datapoints and sequence length. The authors are also not aware of any other MTS dataset that tracks a dynamic system that changes arbitrarily in a largely uncontrolled and inconsistent environment. Furthermore, we demonstrate that this dataset contains temporally distant relationships that previous MTS classification methods struggle with. We hope that the difficulty of this dataset will inspire new and better methods for MTS classification.

We also introduce a more computationally efficient and performant architecture, the Conv-MHSA. This architecture can better capture temporally distant relationships in long sequences and it does so with at much greater computational efficiency than RNN methods. We also show that cutmix, cutout, and mixup augmentations can significantly improve generalization.

The ability to differentiate between pre and post maintenance flights leads can provide a significant benefit to the domain of general aviation. Early detection of maintenance issues has the potential to reduce long term maintenance costs by catching issues before they cause more serious problems. By detecting the need for maintenance one or two days prior to maintenance, we can minimize the amount of flight hours that a pilot spends on compromised aircraft, leading to increased safety. We have already incorporated preliminary models for maintenance classification for NGAFID, which will allow us to gather feedback from users to further refine and improve the early maintenance issue detection system. We hope that these tools will lead to increased safety and reduced costs for general aviation.

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